AIMScribe v3 - Failure Review

Every way the system can fail, what it does today, the evidence, and what to fix next. Reviewed against the code as of 20 October 2026 (recorder 3.5.0, after the second quality round).

Verdict	Count	Meaning
HANDLED	45	Handled, with a test or a live run as evidence
PARTIAL	0	Handled in part, or depends on CMED or the clinic
GAP	0	Not handled: audio can be lost, misfiled or stuck
OUTSIDE	6	Outside our code: needs CMED, the clinic's hardware or how the doctor works - the action and who owns it are listed
Total	51	

What is left, and who acts on it

#	Failure	Consequence	Fix
1. B3	CMED's finished page meets the new recorder	If API 2 is missing for a click, that part is archived unconfirmed (kept)	Run tools/cmed_build_session.py with CMED's build *(CMED)*
2. B1	CMED's server offline (power or internet)	Clinical data missing until CMED sends it	Confirm CMED's outbox in the build session *(CMED)*
3. S5	Recorder not running when the doctor opens the patient	That one consultation is not recorded	See it in the CMED build session (tools/cmed_build_session.py) *(CMED)*
4. E1	Router or WiFi down (CMED's local server unreachable)	Consultations during the outage are not recorded	Ethernet to consulting-room PCs; router and access point on UPS *(clinic)*
5. E2	Power cut at the clinic	As E1 if the router dies	UPS for router and CMED server *(clinic)*
6. S9	Doctor's personal time after a no-prescription visit	Up to 15 min of the doctor's own time in a follow-up's file	Doctors use Pause for personal calls (training); the no-activity limit caps it *(clinic)*

Nothing is left open in our code. Every remaining item needs an action by CMED, the clinic or its IT, listed above with its owner. The most important is the one session with CMED's finished build (B3).

PART 1

Every failure mode, by component

What happens today, with the code it was checked against and the evidence.

Recorder - capture

HANDLED	C1 - Microphone unplugged or headset switched mid-consultation
Today	A watchdog notices within about 2.5 s - failed reads, or the read that never returns when a USB microphone is pulled out - and reopens the Windows default microphone in the same session (core/recorder.py _watch, _failover). The doctor sees 'Microphone changed. Still recording'; the server gets *microphone_switched* with the seconds lost.
Evidence	tests/test_microphone_failover.py (8, fake PortAudio incl. a read that hangs); not tried with a real unplug
Consequence	About 2-3 s of audio at the moment of the switch
Fix	-

HANDLED	C2 - Laptop disk full (C: drive) while recording
Today	A failed clip write no longer stops the segmenter: clips wait in memory, in order, and are written every 3 s once space returns (up to 10 minutes of audio); the power-cut file keeps all of it. *disk_full* is raised once; a new start is refused while the drive has under 1 GB free; free space is checked at start and every heartbeat. A Stop while clips wait answers at once and closes after they are written. Each recording has its own clip writer, so a clip that waited is written into its own recording, never the next patient's (fixed in 3.5.0).
Evidence	tests/test_disk_full.py (15): write failures in clip and journal, order kept, one alert, memory limit, Stop not blocking, chain untouched
Consequence	Audio beyond 10 minutes of a full disk is counted as lost and reported
Fix	-

HANDLED	C3 - Laptop sleeps or the lid is closed mid-consultation
Today	While a recording is open the recorder tells Windows the PC is in use, so idle sleep cannot happen (core/power.py). 'Lid close: do nothing' and USB selective suspend off (which could power the speakerphone down) are set by the separate, signed clinic policy script (recorder/policy), no longer by the installer, so antivirus does not flag the installer.
Evidence	tests/test_power.py; policy script run with -WhatIf; not tried on a real laptop
Consequence	-
Fix	-

HANDLED	C4 - Power cut or forced shutdown mid-recording
Today	The unsealed part of the current clip is written to disk, encrypted, every 3 s (spool partial.aimspl). On the next start it is sealed into the chain as the last segment. Segments themselves are fsynced and renamed; a torn journal line is repaired.
Evidence	tests/test_partial_audio.py (7), tests/test_spool.py
Consequence	At most about 3 s of audio
Fix	-

HANDLED	C5 - Room is silent / microphone muted
Today	Two silent segments raise *silent_session* once; it reaches the dashboard live or from the health log. The clinic microphone is kept at 100% and unmuted (see C7).
Evidence	Live outage test: silent_session arrived from the health log
Consequence	Alert only; a muted session still records nothing
Fix	-

HANDLED	C6 - Windows shutdown, restart or sign-out mid-consultation
Today	A hidden window receives the shutdown notice and pauses the recording, showing 'AIMScribe is saving the consultation' only while one is open. It closes it cleanly (reason *pc_shutdown*) when the shutdown really happens, and resumes it if the shutdown is cancelled (core/shutdown_guard.py). Shutdown is never refused.
Evidence	tests/test_shutdown_guard.py (9, incl. the real hidden window); no real shutdown run
Consequence	-
Fix	-

HANDLED	C7 - Jabra speakerphone not the Windows default, or turned down
Today	The recorder opens the microphone whose name contains 'Jabra' whatever the default is, and keeps its level at 100% and unmuted at start-up and on every idle heartbeat (core/microphone.py). Without it the PC's own microphone records.
Evidence	tests/test_microphone_failover.py (Jabra tests); level read and written live on this PC's microphone; no real Jabra tried
Consequence	-
Fix	-

Recorder - start and labelling

HANDLED	S1 - Wrong patient opened first, then the right one
Today	Every click while the recording is not armed is accepted (RECORDING_STARTED, same recording) and signed into the chain with its audio second. The server applies R3: a first patient open under 60 s, never returned to and never prescribed, belongs to the next. One file under the right patient.
Evidence	Live: Rahim then Karim at 8 s -> one file TEST_KARIM (R3+R1); backend tests/test_labelling.py
Consequence	-
Fix	-

HANDLED	S2 - Recorder crashes after a click, before the close
Today	Click and prescription markers are journalled and delivered in chain order after the restart; the server labels from them as usual.
Evidence	tests/test_start_recovery.py (crash after a click), tests/test_markers.py
Consequence	-
Fix	-

HANDLED	**S3 - Microphone busy at the moment CMED triggers**
Today	Start is kept and retried every 5 s for 15 min until it records, or until CMED sends another patient.
Evidence	tests/test_start_recovery.py
Consequence	Seconds of audio at the start
Fix	-

HANDLED	**S4 - Recorder crashes mid-consultation**
Today	A consultation live within 10 min is resumed after the restart and the new recording names the one it continues (*continues_session_id*). The server checks it is the same PC and patient, confirms it with the first part's API 2, and both archive JSONs carry *continues* / *continued_by*.
Evidence	tests/test_device_state.py, backend tests/test_round2_server.py, archive_worker test_pressure_and_links.py
Consequence	Two linked files instead of one
Fix	-

OUTSIDE	**S5 - Recorder not running when the doctor opens the patient**
Today	The startup task restarts a stopped recorder within a minute. A click made in that minute reaches nothing on our side; whether CMED's page tries again is CMED's code.
Evidence	Installer task: logon trigger repeating every minute + restart on failure
Consequence	That one consultation is not recorded
Fix	See it in the CMED build session (tools/cmed_build_session.py) *(effort: CMED)*

HANDLED	**S6 - Follow-up with no prescription, then the next patient**
Today	The click on the next patient is accepted and logged; the server applies R4: a patient open 60 s or more, not prescribed and not returned from, is a real consultation. The recording is cut at that click into two files.
Evidence	Live: Sakib 70 s, Salma clicked, her prescription -> two files, Sakib 0-66.4 s (R4) and Salma (R1); the two rejoin to the signed hash
Consequence	-
Fix	-

HANDLED	**S7 - Recording never stopped**
Today	A recording ends at the next click after a prescription; 2 min (+ up to 10 s, at a pause) after the prescription; after 15 min with no CMED activity; at Windows shutdown; or with Stop.
Evidence	Live: closed by itself 121 s after the prescription (counselling_limit); tests/test_markers.py (both limits)
Consequence	-
Fix	-

HANDLED	**S8 - Same trigger sent twice / page reload**
Today	Before the prescription: accepted, same recording, no extra marker. After it: SESSION_ALREADY_ACTIVE. A repeated prescription_built: GATE_ALREADY_ARMED.
Evidence	tests/test_channel_a.py, tools/channel_a_test.py 24/24
Consequence	-
Fix	-

OUTSIDE	S9 - Doctor's personal time after a no-prescription visit
Today	With no prescription there is no signal for the moment the patient left, and no software can tell a phone call in the room from a consultation. Until the next click, or 15 min without CMED activity, it is recorded into that patient's file.
Evidence	By design
Consequence	Up to 15 min of the doctor's own time in a follow-up's file
Fix	Doctors use Pause for personal calls (training); the no-activity limit caps it *(effort: clinic)*

HANDLED	S10 - Cases the rules cannot settle
Today	Kept, labelled by the best rule and flagged; the dashboard's **Needs review** panel lists every *labelling_review* and *relabelled_after_archive* case with its parts, rule and file names.
Evidence	backend tests/test_labelling.py, tests/test_round2_server.py (review panel)
Consequence	A person looks at the flagged file from the dashboard
Fix	-

HANDLED	S11 - API 3b arrives after the files were archived
Today	The rules run again. If the cut changes, the archive rejoins the parts, checks the result against the signed hash, cuts again and replaces the files; *relabelled_after_archive* is raised.
Evidence	archive_worker test_parts.py (recut, matching and mismatching hash); tests/test_labelling_server.py
Consequence	-
Fix	-

Recorder to server

HANDLED	U1 - Laptop clock wrong (behind or ahead of the server)
Today	The recorder measures the server's time from the Date header on every reply and checks each grant against it (core/uploader.py note_server_date, server_now; core/crypto.py verify_grant now=). Used grants are remembered by when this PC saw them, not by their expiry, so a fast clock cannot accept one twice. The offset goes in the heartbeat; the server raises *clock_skew* once per 6 h when it is 2 minutes or more. The installer turns on Windows time sync.
Evidence	tests/test_clock_skew.py (6 s, 5 min and 1 h either way), backend tests/test_clock_skew_server.py; Channel A 24/24 on the new build. Not tried on a laptop with its clock actually set wrong
Consequence	Times stamped by the PC (and so file names) follow its own clock until time sync corrects it
Fix	-

HANDLED	U2 - Internet down for hours
Today	Recording never waits; spool holds 13.5 h; authorisation retried; uploads and matching resume on reconnect.
Evidence	Live outage tests (60 s and 100 s offline) -> archived
Consequence	Delayed only
Fix	-

HANDLED	**U3 - Server says no (unmapped clinic, revoked device, key mismatch)**
Today	Held on the laptop, asked again every 5 min, nothing deleted. Each heartbeat now carries the held recordings and why; the dashboard shows them per PC (**Held on PCs** panel and the Held column).
Evidence	tests/test_device_state.py, tests/test_round2_server.py
Consequence	Delayed until the cause is fixed; visible on the dashboard
Fix	-

HANDLED	**U4 - What the PC saw while offline**
Today	Offline heartbeats (1/min) and every alert go to a 5 MB health log, delivered on reconnect and deleted only after the server stores them.
Evidence	Live outage test: 3 lines delivered then deleted; dashboard showed the period
Consequence	-
Fix	-

HANDLED	**U5 - Duplicate upload after a lost response**
Today	Server recognises the stored entry hash and answers duplicate.
Evidence	Segment, reassign and marker duplicate tests
Consequence	-
Fix	-

HANDLED	**U6 - A clicked patient's grant refused or invalid**
Today	The marker is still delivered (grant_jti null after the hold period) and kept in the chain; the server finds that patient's API 2 and 3b by the click's five fields instead (*marker_without_grant* / *marker_grant_invalid* alerts).
Evidence	tests/test_markers.py (grant cases), tests/test_labelling_server.py
Consequence	-
Fix	-

HANDLED	**U7 - PC installed while the clinic's internet is down**
Today	The enrolment token is tried again every minute until the server answers; a token the server refuses is not retried.
Evidence	tests/test_enrol_retry.py
Consequence	-
Fix	-

Server and database

HANDLED	D1 - A migration not applied on a deployed database
Today	At start-up the server checks for what each migration 002-015 and the clinical schema leave behind - tables, columns, the entry types the chain accepts - so a skipped or half-applied migration is named (backend/src/schema_check.py). It is logged as critical, raised as *schema_missing*, shown in /health ("schema": "missing 015_...", status degraded) and alerted by the monitor. It clears by itself within a minute of applying the file. AIMS_SCHEMA_STRICT=true refuses to start instead.
Evidence	tests/test_schema_check.py (9, real PostgreSQL, each migration removed in turn); deploy/uiu tests (monitor); live: local server reports schema ok
Consequence	-
Fix	-

HANDLED	D2 - Database restarts mid-request
Today	The request fails; the recorder and worker retry; commits are idempotent.
Evidence	Duplicate-detection tests; not re-run against a restarting database
Consequence	Delayed only
Fix	-

HANDLED	D3 - API 2 never arrives for a recording
Today	Recording archived 'unconfirmed' and kept; API 2 attaches whenever it arrives.
Evidence	Live: earlier TEST_NEXT and TEST_OFF archived unconfirmed
Consequence	-
Fix	-

HANDLED	D4 - A CMED outage makes many recordings unconfirmed
Today	One *unconfirmed_recording* alert per clinic per 6 hours, carrying the count and the recordings, instead of one per recording.
Evidence	tests/test_round2_server.py
Consequence	-
Fix	-

HANDLED	D5 - Background audit writer stops
Today	Restarted on demand when dead or bound to a closed loop.
Evidence	441 backend tests pass against real PostgreSQL
Consequence	-
Fix	-

HANDLED	D6 - The wrong patient's late records reach a relabelled file
Today	A visit finds a recording only through the grant that recording now carries, or one of its clicks; a first-patient grant left behind by a move confirms and attaches nothing.
Evidence	tests/test_reassign.py, tests/test_labelling_server.py
Consequence	-
Fix	-

Archive, storage and backups

HANDLED	A1 - Server disk fills
Today	When the archive disk falls below its floor, the worker releases the oldest local WAVs that have a proven cloud copy and still match their hash, before the retention period, until there is room; never one without a verified copy. Copies overdue more than 2 days already alert.
Evidence	archive_worker test_pressure_and_links.py (13)
Consequence	If R2 itself is down for weeks, the disk can still fill - and the monitor says so
Fix	-

HANDLED	A2 - R2 unreachable or wrong keys
Today	/health reports the bucket state; copies are retried; *cloud_copy_overdue* after 24 h.
Evidence	tests/test_health_storage.py
Consequence	Copies delayed
Fix	-

HANDLED	A3 - The droplet is lost
Today	Audio is safe in R2. Each night's encrypted database backup is now also copied to its own R2 bucket with its own token, size-checked after upload, kept 90 days, and caught up the next night if R2 could not be reached; unencrypted backups never leave (deploy/uiu/backup/backup.sh). *fetch latest* brings one back for restore.sh. The monitor alerts when last night's backup, or its off-site copy, is over 36 h old.
Evidence	deploy/uiu tests/test_backup_offsite.py (real backup.sh, stand-in rclone); live: the real image dumped, encrypted and copied both databases to a test bucket, the copy was fetched back and restored with all 10 sessions
Consequence	At most one night of clinical records, if the droplet is lost before 02:30
Fix	-

HANDLED	A4 - Archive worker dies mid-copy or mid-cut
Today	Partial files are discarded and the work restarts; a multi-part cut resumes; pieces deleted only after every part is copied and verified.
Evidence	archive_worker test_parts.py, test_worker_copy.py
Consequence	-
Fix	-

HANDLED	A5 - A cut file does not match the signed recording
Today	Parts are contiguous and checked to add up to every frame; rejoined, they must equal the session's signed hash before any re-cut, or the worker refuses and reports.
Evidence	Live: Sakib + Salma rejoined to the session's sha256 exactly; test_parts.py
Consequence	-
Fix	-

CMED boundary

OUTSIDE	B1 - CMED's server offline (power or internet)
Today	API 2 and API 3 must wait in CMED's own queue; recordings wait on our server and are confirmed whenever either arrives. A clinic CMED is silent for erases nothing; only a recording CMED never confirms is erased, 7 days after it was made.
Evidence	Depends on CMED's queue
Consequence	Clinical data missing until CMED sends it
Fix	Confirm CMED's outbox in the build session *(effort: CMED)*

HANDLED	B2 - Same visit or prescription sent again
Today	Identical: 200 ALREADY_RECEIVED; changed prescription: next version; previous visit fingerprinted (ALREADY_STORED / UPDATED / STORED).
Evidence	tests/test_previous_visit.py
Consequence	-
Fix	-

OUTSIDE	B3 - CMED's finished page meets the new recorder
Today	Every reply is one their code already treats as success; nothing new is asked of CMED. Not yet seen: whether their page sends API 2 for every click and API 1 again on return.
Evidence	Our reference page and /manual-test; tools/cmed_build_session.py ready
Consequence	If API 2 is missing for a click, that part is archived unconfirmed (kept)
Fix	Run tools/cmed_build_session.py with CMED's build *(effort: CMED)*

HANDLED	B4 - Unmapped or wrong clinic code
Today	Channel B: stored with an *unmapped_clinic* alert. Recorder: CLINIC_MISMATCH, held.
Evidence	tests/test_v3_server.py
Consequence	Delayed until mapped
Fix	-

Security

HANDLED	X1 - Laptop stolen
Today	Spool and partial audio are encrypted with a DPAPI-wrapped key; the device can be revoked. Each recorder now reports whether its drive is BitLocker-protected; the dashboard shows it and raises *disk_not_encrypted* once a day for a PC without it.
Evidence	tests/test_disk_full.py (BitLocker probe), tests/test_round2_server.py; live: this PC reported not encrypted
Consequence	Turning BitLocker on is the clinic's IT
Fix	-

HANDLED	X2 - One hospital's CMED key leaked
Today	That hospital only is exposed: each hospital has its own key pair and 5-minute tokens. Revoke that hospital's client (POST /api/v2/admin/cmed-client/{id}/revoke) - its keys and every live token stop at once; the other hospitals are unaffected.
Evidence	backend tests/test_cmed_auth_fhir.py, test_cmed_hardening.py (revoke, key revocation ends its tokens)
Consequence	-
Fix	-

HANDLED	X3 - Forged or replayed grant / chain entry
Today	EdDSA only, single-use grants, signed hash chain verified per entry, markers included. Each entry must name the clip it covers and follow the head exactly; every device request is signed with the PC's key (single-use nonce, 5-minute window).
Evidence	tests/test_authorisation.py, test_chain.py, backend tests/test_data_safety.py
Consequence	-
Fix	-

Deployment and versions

HANDLED	V1 - New recorder against a server without the new endpoints
Today	The recorder keeps its audio and retries; the server's start-up check names any missing migration in the log, /health and an alert, and the runbook orders server first.
Evidence	tests/test_schema_check.py; live: schema_missing raised when the server started without 016
Consequence	Delayed until the server is upgraded
Fix	-

HANDLED	V2 - Old recorder builds still installed at clinics
Today	Every heartbeat carries the version; the dashboard flags PCs below AIMS_MIN_RECORDER_VERSION (3.4.0; raise it to 3.5.0 once the signed installer is rolled out) and raises *recorder_outdated* once a day.
Evidence	tests/test_round2_server.py; live: the 3.3.0 build was flagged, 3.4.0 was not
Consequence	Reinstall flagged PCs
Fix	-

HANDLED	**V3 - Recorder keys do not match the new server**
Today	Recordings are held, not deleted; tools/check_key_pair.py and runbook step 11b check it.
Evidence	Key-pair check
Consequence	Delayed until fixed
Fix	-

HANDLED	**V4 - Upgrading a PC with the installer**
Today	The installer keeps that PC's server address, CMED address and local key; the server address is set once at build time. Unused enrolment tokens last 14 days.
Evidence	Installer compiled; tools tested; upgrade not run on a real PC
Consequence	-
Fix	-

Clinic environment

OUTSIDE	**E1 - Router or WiFi down (CMED's local server unreachable)**
Today	The doctor cannot open a patient, so no trigger and no recording; a session already running keeps recording.
Evidence	By design (no manual start)
Consequence	Consultations during the outage are not recorded
Fix	Ethernet to consulting-room PCs; router and access point on UPS *(effort: clinic)*

OUTSIDE	**E2 - Power cut at the clinic**
Today	A laptop on battery keeps recording; a hard power-off loses at most about 3 s; server and router need UPS (see E1).
Evidence	Live outage tests; tests/test_partial_audio.py
Consequence	As E1 if the router dies
Fix	UPS for router and CMED server *(effort: clinic)*

HANDLED	**E3 - Windows update restarts the PC mid-consultation**
Today	The installer sets Windows Update active hours 07:00-23:00 and no automatic restart while someone is signed in. If a restart still happens, the recording closes cleanly (C6) and the consultation resumes as a linked second part (S4).
Evidence	Installer compiled; shutdown and resume tests
Consequence	-
Fix	-

HANDLED	**E4 - 28 rooms opening at once**
Today	Grant path moved off shared locks; measured earlier under the 14- and 28-room simulations.
Evidence	Earlier simulations; not re-run against the labelling changes
Consequence	-
Fix	-

PART 2

Recommended next fixes, in order

Lost or misfiled audio first, then anything that stalls silently.

Order	Item	What
1	B3/B1/S5	One session with CMED's finished build: tools/cmed_build_session.py
2	E1/E2	Clinic set-up: Ethernet and UPS for the consulting-room PCs and the router
3	S9	Doctor training: Pause for personal calls
4	X1	Clinic IT: BitLocker on each PC (the dashboard shows which have it)
5	V2	Roll out the signed 3.5.0 installer to every PC, then set AIMS_MIN_RECORDER_VERSION=3.5.0

Not verified in this review

• A real USB microphone pulled out (C1), a real Jabra (C7), a real Windows shutdown (C6), a hard power cut (C4) and a real lid close (C3): each is tested through the same code with a fake device or message, not on clinic hardware.
• CMED's finished page (B3): tools/cmed_build_session.py is ready, not yet run.
• 28-room load (E4) and database restart (D2) were not re-run against today's code.
• Router, UPS and Windows-update behaviour (E1-E3) depend on each clinic's set-up.